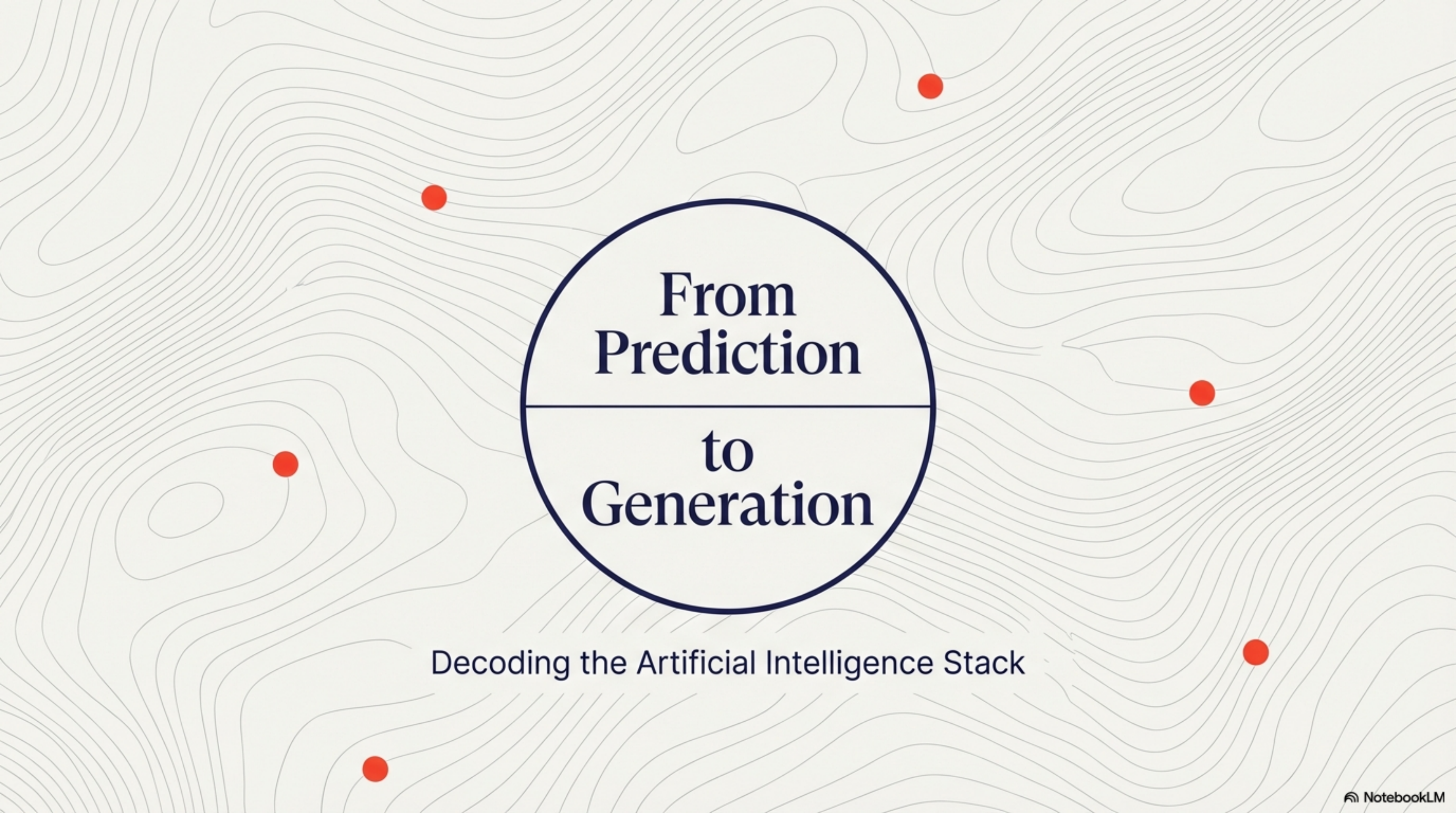


The background is a light gray topographic map with intricate, wavy contour lines. A large, dark blue circle is centered on the page, divided horizontally by a thin dark blue line. Five solid red dots are scattered across the map: one in the upper right, one in the upper left, one on the left side, one on the right side, and one in the lower left.

# From Prediction to Generation

Decoding the Artificial Intelligence Stack



# The Fundamental Purpose of Predictive AI

Millions of Historical Records + Current Conditions

## Navigation Context

**Data:** Traffic density, distances, modes of transport, millions of previous trips.

**Query:** Adyar to Velacherry, 9 AM, driving a car.

**Output:** Exact time and shortest route with 90-95% accuracy.

## Meteorological Context

**Data:** Billions of historical weather points from March 1990, 2024, 2025.

**Query:** Today's conditions.

**Output:** Exact prediction of 30 degrees Celsius.



← The Prediction:  
A Single Accurate Answer



# The Concentric Universe of Machine Intelligence

## Artificial Intelligence (AI)

Tasks done by machines that mimic human behavior.

## Deep Learning (DL)

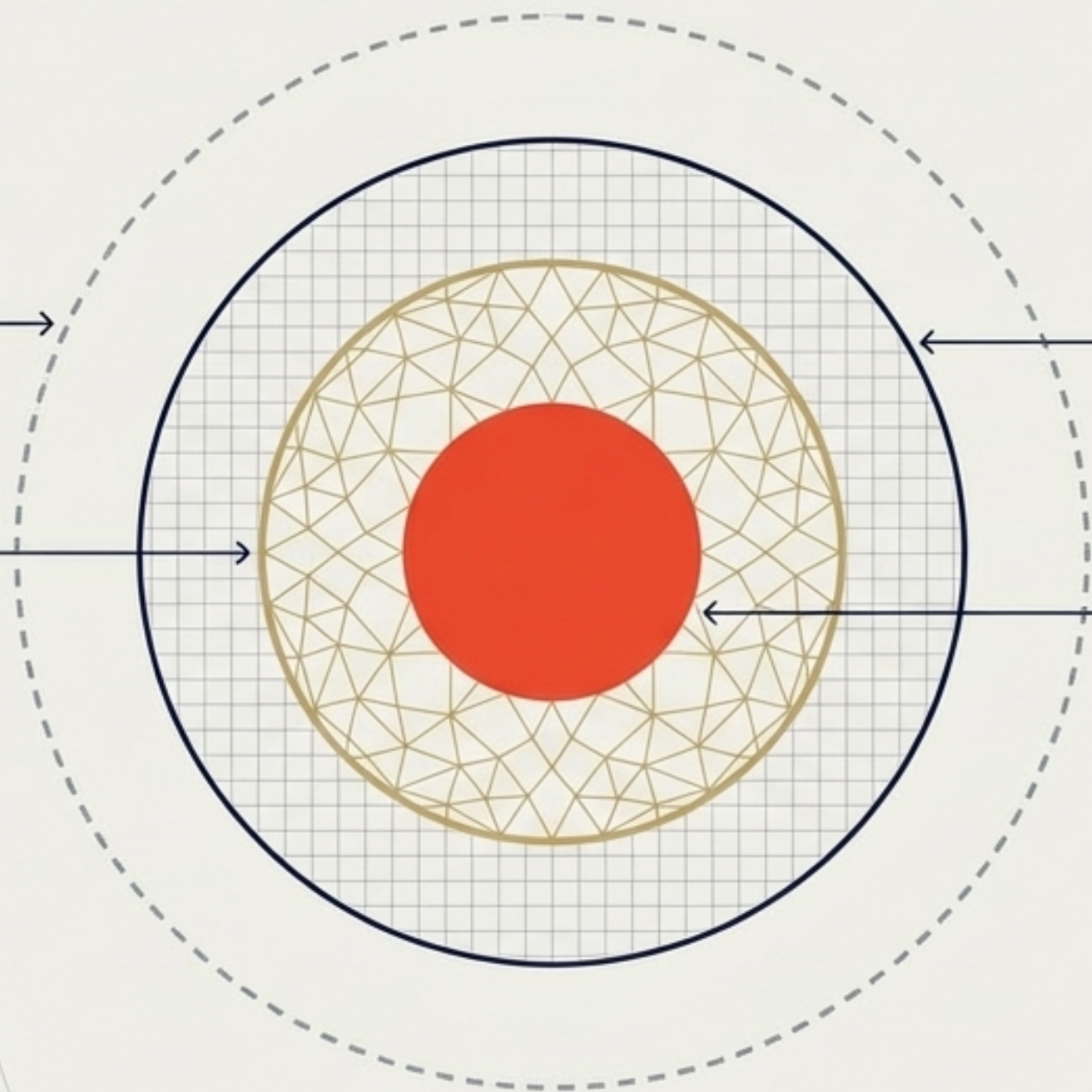
Vast data models powered by scale and Neural Networks.

## Machine Learning (ML)

Predictive models trained on thousands to millions of records.

## Generative AI (Gen AI)

Content-generating models that create novel outputs.





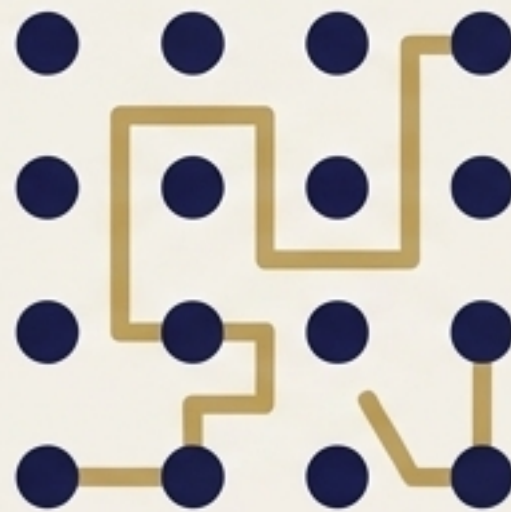
# Machine Learning Scales Pattern Recognition

## 1. Structured Inputs



Operates on structured records ranging from 1,000 to millions of entries.

## 2. Pattern Identification



A dedicated subset of AI focused entirely on analyzing historical patterns.

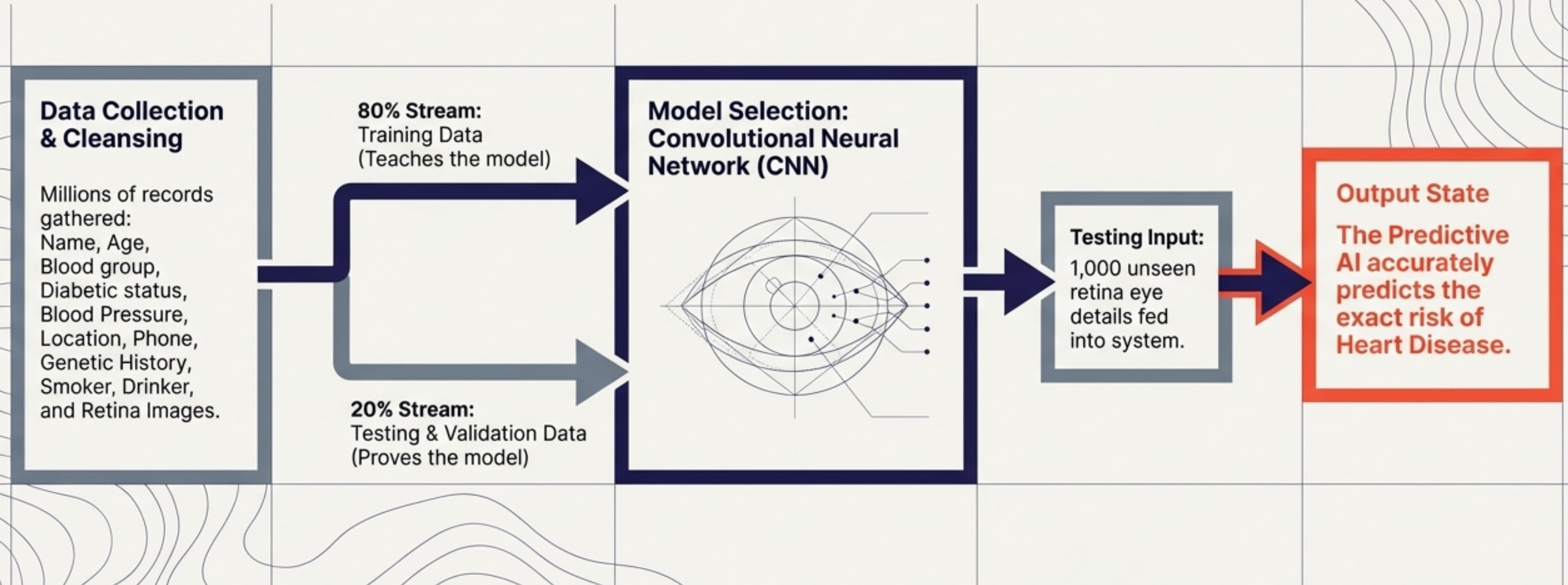
## 3. Definitive Output



The sole output is a definitive, highly accurate prediction based on the data.



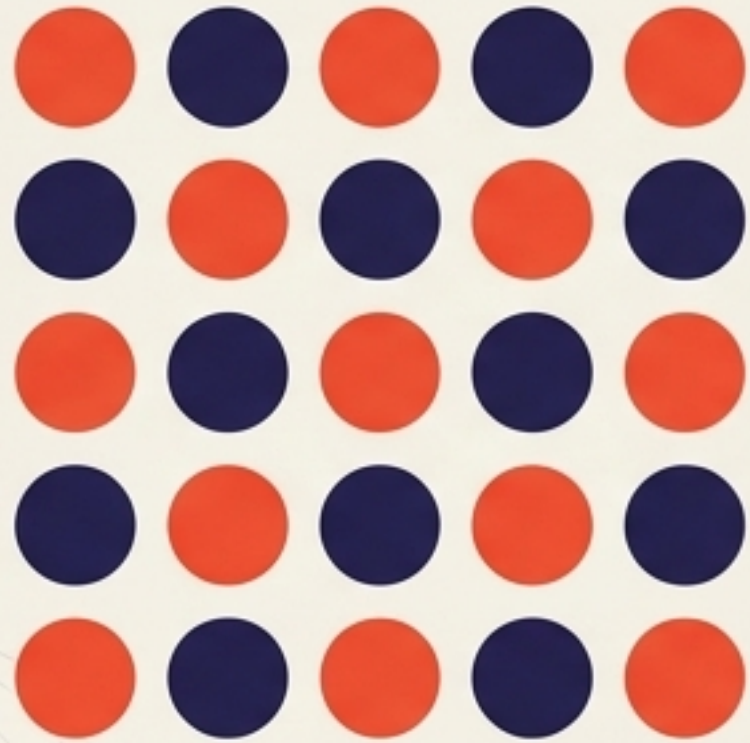
# The Predictive Pipeline: From Raw Data to Diagnosis





# Mechanism and Intent: Supervised vs. Unsupervised Learning

## Supervised Learning (Guiding)

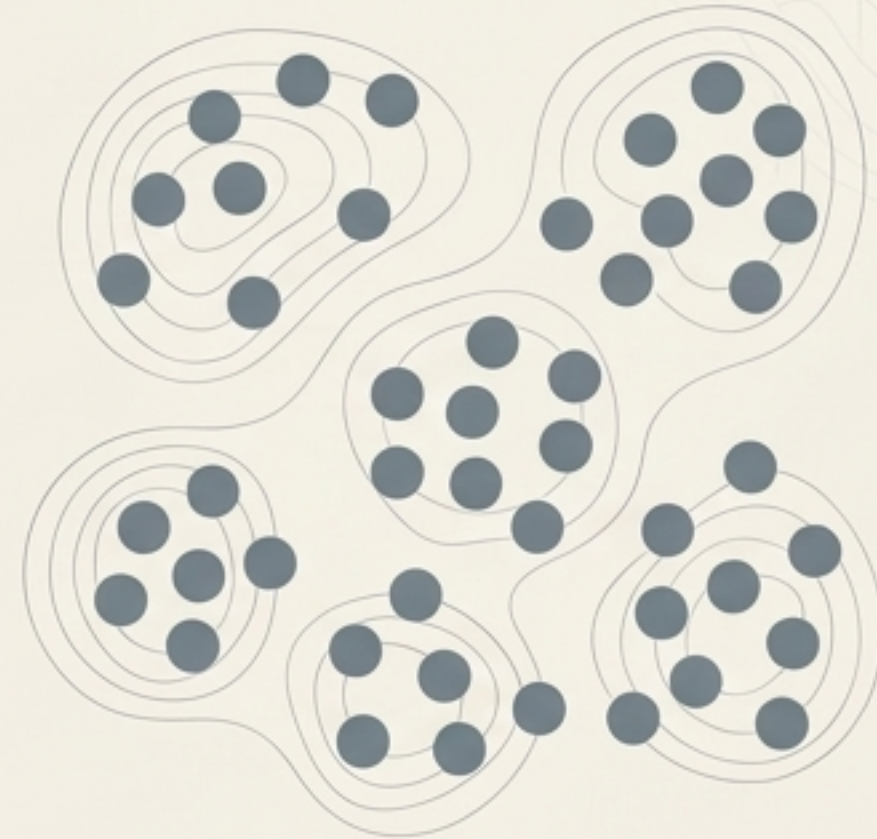


**Input Requirement:** Labelled Data.

**Mechanism:** Content-based learning with explicit targets (e.g., Star ratings, explicit likes/dislikes).

**Goal:** Train the model to map inputs to known, predefined outputs.

## Unsupervised Learning (Discovering)



**Input Requirement:** Unlabelled Data.

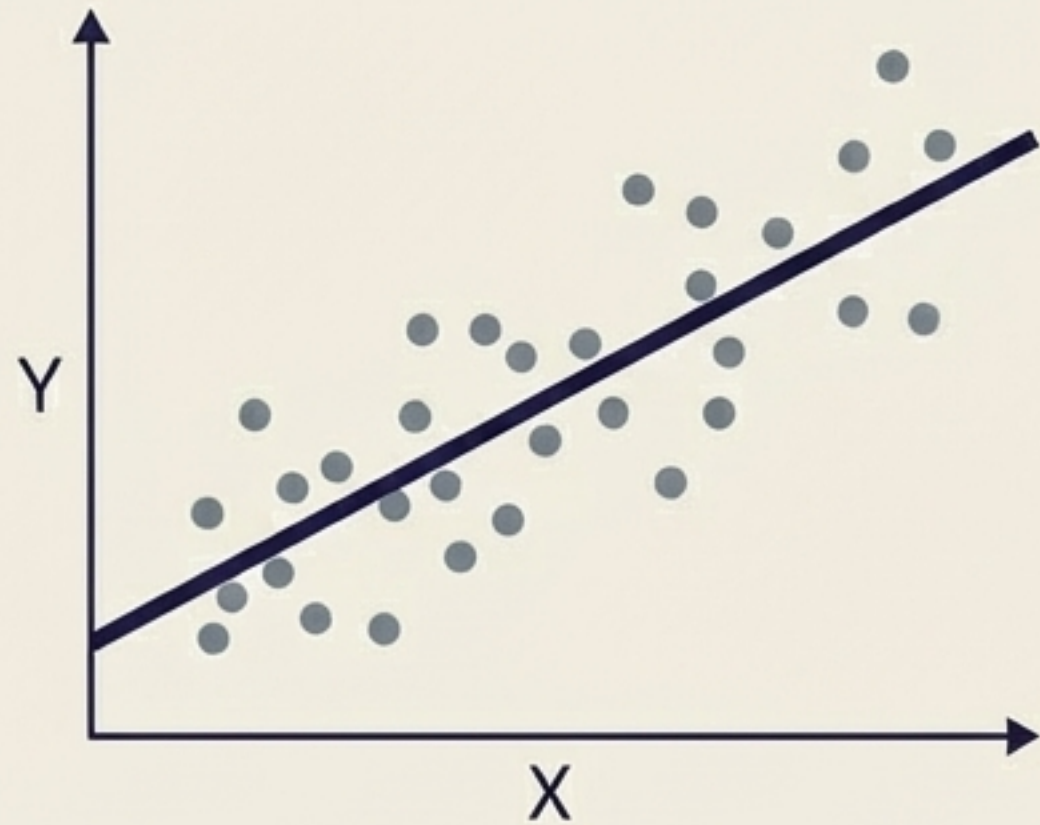
**Mechanism:** Behavior and genre analysis without explicit tags (e.g., Collaborative filtering without likes or comments).

**Goal:** Discover hidden structures, patterns, or groupings within raw data.



# Visualizing Supervised Outputs: Numbers vs. Categories

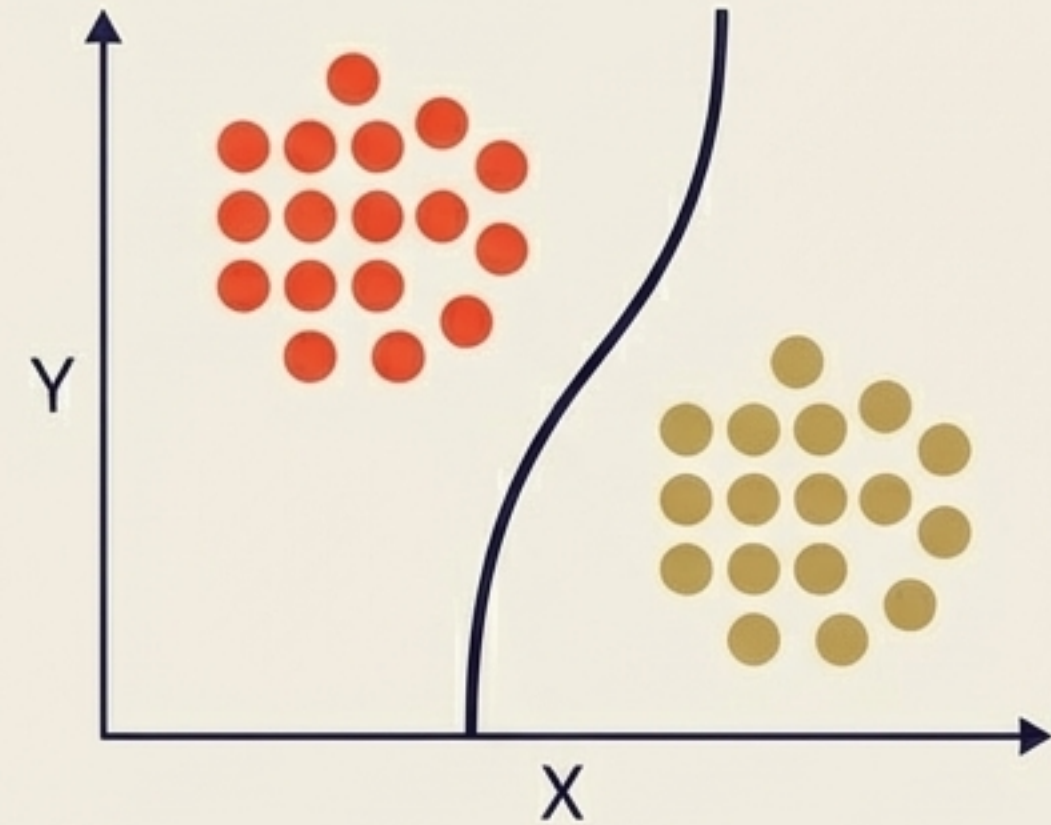
Linear Regression (Numerical Output)



**Definition:** Predicts a specific, continuous number.

**Real-World Use Cases:** Forecasting Real Estate home purchase prices; predicting the exact number of defects in software testing.

Classification (Categorical Output)



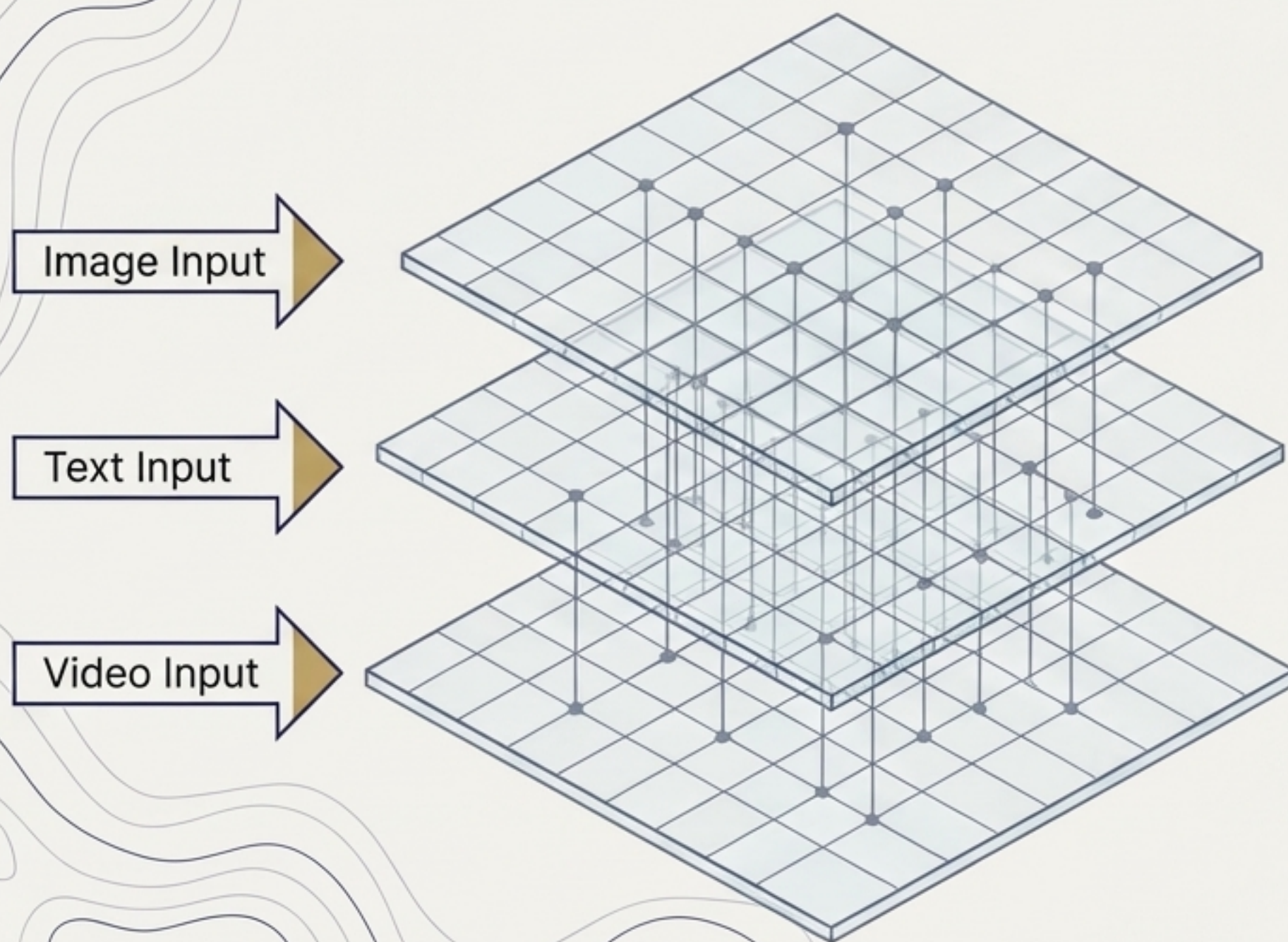
**Definition:** Sorts inputs into distinct predefined labels or categories.

**Real-World Use Cases:** Google predicting the risk of a heart attack (Output: strictly Yes or No); Defect classification routing.



# Scaling the Architecture: Deep Learning

Deep Learning = Predictive ML + Trillions of Records + Neural Networks



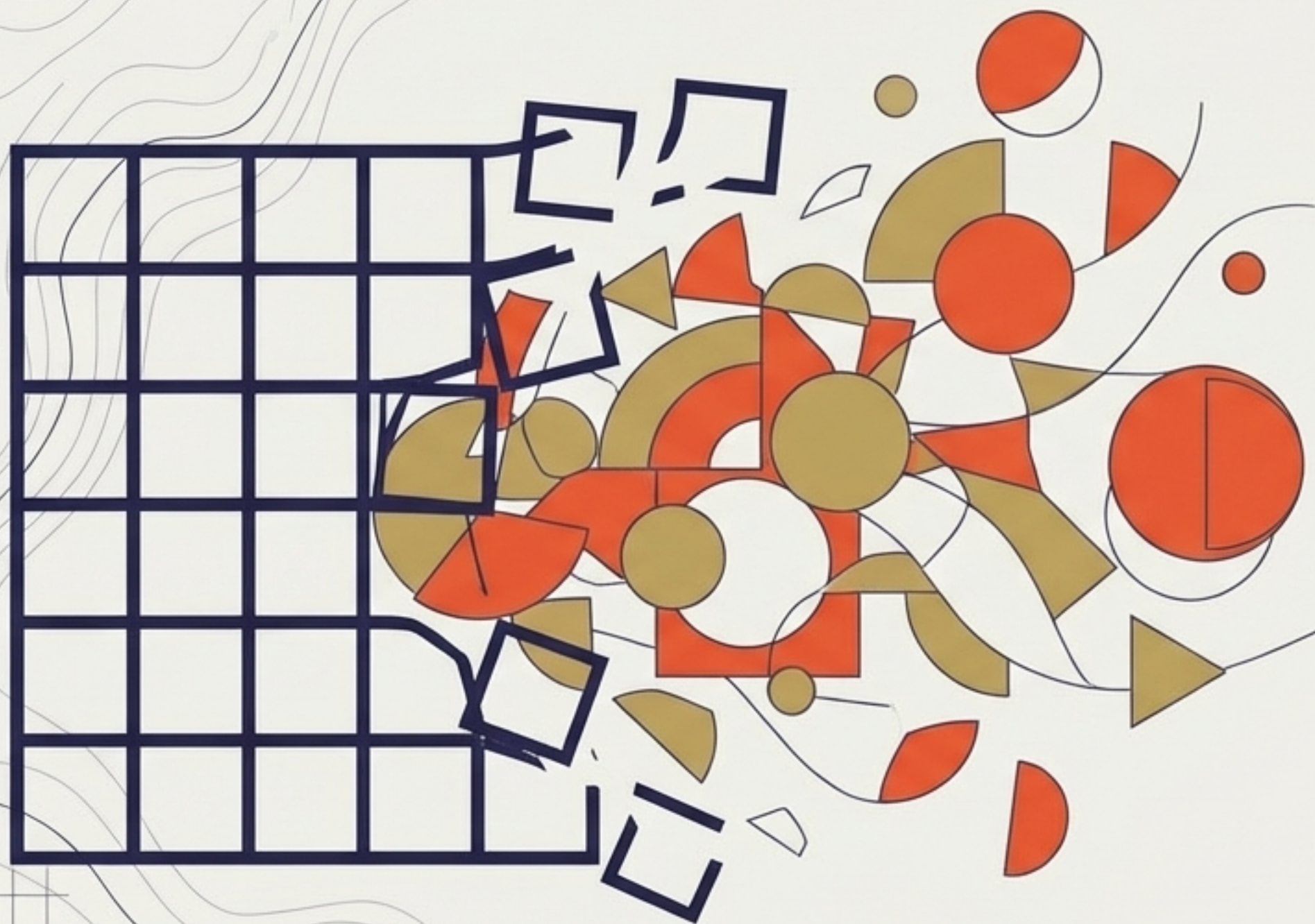
A subset of Machine Learning designed for extreme scale.

Operates on millions to trillions of records.

Multimodal Capabilities:  
Unlike basic ML, Deep Learning seamlessly accepts and processes diverse inputs simultaneously.



# The Paradigm Shift: Generative AI



A specialized subset of Deep Learning that is **strictly non-predictive**.

**The Shift:** Instead of analyzing vast amounts of data to find a single existing answer, its ultimate output is to Generate The Content.

**Omni-Domain Capabilities:**

Text | Image | Audio | Video | Code

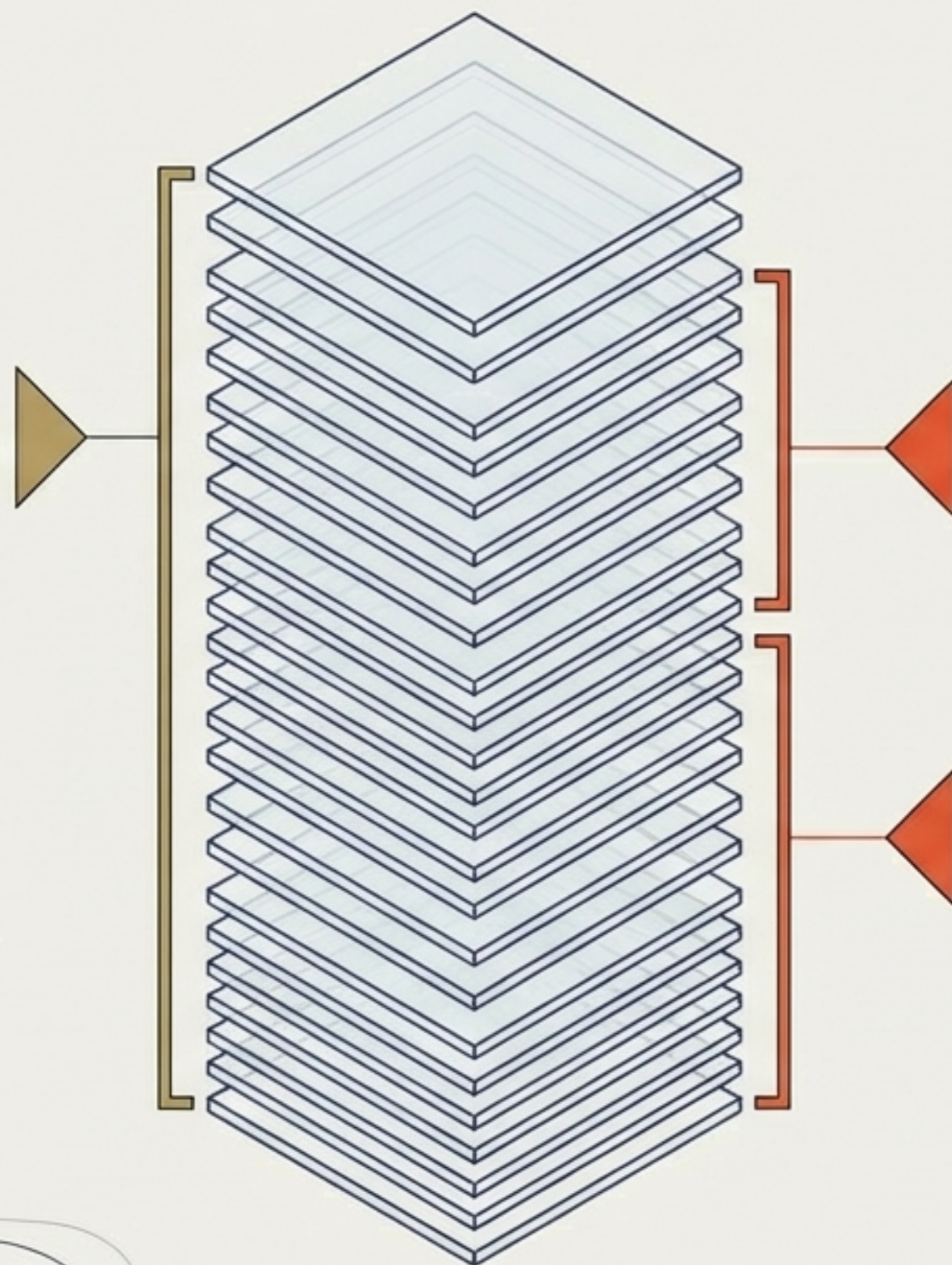


# The Engine of Creation: Large Language Models (LLMs)

## GPT (Generative Pretrained Transformer)

Pretrained on massive datasets encompassing all fields, vastly exceeding just software or technical domains.

**Service Provider:** OpenAI  
(Creators of ChatGPT)



## GPT-4 Architecture

Engineered with **96 to 120** internal layers.

## GPT-5 (LLM 5.3/5.4) Scale

Scaling up to **190** internal layers and an unprecedented **120 Billion** parameters.



# The Synthesis: Predictive vs. Generative Intelligence

|                  | Predictive AI (ML/DL)                                  | Generative AI (LLMs)                       |
|------------------|--------------------------------------------------------|--------------------------------------------|
| Primary Goal     | Accurate Single Answer                                 | Novel Creation                             |
| Data Input Scale | Thousands to Trillions of structured records           | Internet-scale, multi-domain pretraining   |
| Output Type      | Numbers, Categories, Routes, Medical Diagnoses         | Generated Text, Code, Images, Audio, Video |
| Prime Example    | Google Maps Routing; CNN Heart Disease Risk Prediction | OpenAI's ChatGPT (GPT-4 / GPT-5)           |



# Day 2 Preview: Inside the Black Box

Tomorrow, we dissect the anatomy of the Large Language Model and explore its internal workings in unprecedented detail.

